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CRITICAL CARE

Management of Pain, Agitation, Delirium, Immobility and Sleep (PADIS)

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Key Points

- YG Critical Care uses the PAD (Pain, Agitation, Delirium) model. Assess and treat (in turn) pain, then sedation, then delirium.
- Sedated patients should be assessed for pain using the CPOT score
- Opiate and specific treatment options are described.
- First Line sedation is propofol. Monitor for Propofol Related Infusion Syndrome.
- To diagnose delirium, patients must have an altered mental state **PLUS** inattention **PLUS** either an altered level of consciousness **OR** disorganised thinking (As per YG CG Assessment of Delirium).
- Staff can access '000 NHS – Pain, Agitation & Delirium' module on ESR.

Introduction

Pain, agitation and delirium are common in critical care, affecting up to 50%, 70%[3], and 33% of patients respectively. Distress on the unit can be from one or more of the triad, but for appropriate treatment, the underlying problem must be diagnosed. We follow the PAD model, as recommended by the Intensive Care Society and the all-Wales online guidance. This has been expanded to give local prescribing advice. Staff are recommended to undertake '000 NHS – Pain, Agitation & Delirium' on ESR. International guidelines include early rehab (Immobility) and Sleep to make a PADIS model. Other than to say early rehab with mobilisation and sleep are important, including avoidance of delirium.

Step 1: Pain

50% of patients suffer pain on ICU, e.g. from repositioning, deep suction, surgical wounds and line insertion. This has important sequelae such as aggression, asynchrony and long term cognitive impairment. Pain assessment should happen at least once a shift, but is recommended to occur every 4 hours.

For awake patients, we recommend asking the patient to rate pain out of 10. If non-verbal, consider using the VAS (Visual Analogue Scale) or Wong-Baker FACES pain score. Scores of 1-3 are mild, 4-6 are moderate, ≥7 is severe.

In sedated patients, pain assessment is difficult. Vital signs do not correlate with self-reported pain scores, and cannot be used as a substitute. We recommend the CPOT (Critical Care Pain Observation Tool). For the CPOT, observe the patient at rest for 1 minute, and also during care interventions such as deep suctioning or repositioning. Attribute the highest score for each domain seen during the assessment period.

A score of >2 suggests unacceptable levels of pain, and should result in an increase in analgesia. Pain prescriptions should be re-assessed daily, taking into account renal function, allergies and twice-daily pain scores. Consideration should be given to administering pre-procedural analgesia.

Pain Management – Pharmacological Treatment.

- All Patients with acute pain should have **paracetamol** 1g QDS, unless weight <60kg or severe hepatic impairment (raised ALT with impaired synthetic markers such as INR and albumin).
- For Un-intubated patients with normal absorption, use titrated doses of **Oramorph 5-15mg Hourly**, and once requirements known, prescribe 2/3^{rds} of daily requirements as **Zomorph**, split into BD dosing.
- For those with renal impairment (eGFR <60) consider **oxynorm** 1.25-5mg hourly and **oxycontin** as long acting alternative.
- For those with poor absorption, prescribe **Morphine or Fentanyl PCA** or Alfentanyl infusion as per ICU drug chart; analgesia and sedation page.
- For sedated patients pres **Alfentanyl** as per ICU drug chart, analgesia and sedation page.
- Consider **Gabapentin** (titrated as per BNF) for neuropathic pain.
- For post-surgical cases where opioid reduction is desirable, consider ketamine (0.5 mg/kg IV bolus x 1, followed by 1-2 µg/kg/min infusion)[1].
- Consider **Ibuprofen** and/or regional (epidural or ESP) for rib fractures. Discussion with a consultant is required for these, guided by local 'Management of Patients with Chest Wall Injuries' guideline.

Step 2: Agitation and Sedation Practice

Sedation can be used to aid tolerability of distressing interventions on ICU, primarily ventilation. It should not be used to treat pain, see above. There are specific indications for deep sedation:

- Status Epilepticus
- Raised Intracranial Pressure – e.g. traumatic brain Injury or Sub-arachnoid Haemorrhage
- Severe Respiratory Failure – with or without neuromuscular blockade.

Outside of these specific indications, light sedation should be targeted. This will promote liberation from respiratory support, avoid psychological complications and aid neurological assessment.
We use the Richmond Agitation and Sedation Score (RASS) see below. RASS target for light sedation is -2 to 0. For scores of +1 to +4 sedation should be increased. For scores -3 to -5, sedation should be held until target range achieved, then re-started at half the original rate.
Daily sedation holds should be used in addition to targeted light sedation.[1]
Patients requiring neuro-muscular blockade should have BIS monitoring, with a target of 40-60.

Sedation Regimes

1% Propofol is the first line sedative. If infusion rates exceed 20ml/hr, please use 2% to limit lipid infusion. Although it can occur at any dose, dose must not exceed 4mg/kg/hr, due to the higher risk of propofol related infusion syndrome (PRIS), see below. Second line, including in non-alcohol substance misuse, is clonidine infusion (750mcg in 50mls normal saline, 0-5mls/hr, observe for bradycardia and hypotension).

For patients with a history of alcohol abuse, consider midazolam infusion (50mg midazolam in 50mls, rate 0-5mls/hr). This is the only routine use of benzodiazepines in critical care sedation, and should otherwise be avoided due to the risk of precipitating delirium.

For patients with severe bronchospasm, substance misuse, or difficulty in sedating, consider use of the 'sedaconda' to administer isoflurane. See separate guidance.

Propofol Related Infusion Syndrome – PRIS

PRIS occurs in 1% of general ICU patients, but rates are much higher if 4mg/kg/hr is infused. PRIS carries an 18% mortality but is often missed. It presents as:

- Cardiac: Bradycardia, Ventricular arrhythmia, Cardiac Failure, ECG changes (widened QRS and Brugada), asystole.
- Hepatic: Elevated transaminases and steatosis.
- Metabolic: Lactaemia, Acidosis, Hyperkalaemia, Hyperlipidaemia
- Renal/muscle: myopathy, Rhabdomyolysis, AKI

Diagnostic criteria are:

1. metabolic acidosis plus
2. cardiac dysfunction
3. and at least one other feature:
    a. Hypertriglyceridemia
    b. Rhabdomyolysis
    c. Acute kidney injury occurred after the start of propofol.

Assessment and treatment of PRIS

- Prevent – Consider alternative sedation
- Screening: CK <5000 is low risk marker (<1%)
- Stop propofol
- Supportive therapy - No specific antidote
    - Fluid Resus and treat Hyperkalaemia
    - Treat bradycardia, but may be resistant to drugs and TC Pacing
    - Consider treatment with milrinone, glucagon and calcium
    - HF or HD to eliminate propofol and metabolites, and treat acidosis and AKI, +/- NaHCO3
    - ECMO has been used effectively

Step 3: Delirium

Delirium occurs in 1/3 of critical care patients. It causes increased increased hospital length of stay and long term (12month) cognitive impairment. The mean duration of delirium is 2-3 days (Inter-quartile Range 2-7days).

Screening

All patients admitted to ICU are at risk of delirium

- Screen for delirium within 24 hours of admission using the CAM-ICU assessment tool.
- Repeat the screening every 12-24 hours or sooner if a concern regarding delirium arises.
- Document clearly in the patient's notes.

Risk factors

The single greatest modifiable risk factor is the use of benzodiazepines, with 4% rise in incidence with every 5mg of benzodiazepine administered. Benzodiazepines and 'Z' drugs such as zopiclone should be avoided wherever possible, outside of

Drug side effects and withdrawal

Electrolyte imbalance

Liver, cardiac or respiratory failure

Infection/infarction

Retention of urine/faeces

Intracranial event

Uraemia due to dehydration

Metabolic

seizure activity and alcohol withdrawal. Blood transfusion is the only other modifiable risk factor, and a usual transfusion threshold of 70mg/dL should be used.[1]

In a systematic review done in 2015, Zaal et al showed 11 risk factors are supported by strong or moderate evidence. These include:

Age (>65yrs), pre-existing cognitive impairment (e.g. dementia), previous delirium, Hypertension, reduced conscious level/coma, acute illness, co-morbidities, emergency surgery, mechanical ventilation, organ failure, poly-trauma, metabolic acidosis.

Other factors may include drugs or alcohol dependence and nicotine dependence.

Diagnosis

- Diagnose using the CAM-ICU tool
    - Altered mental state ***plus*** Inattention ***plus EITHER*** altered level of consciousness ***or*** disorganised thinking.

Prevention and management by non-pharmacological means

- Ensure regular patient engagement and include orientation to time and place and reassurance.
- If the patient usually has glasses and/or hearing aids make sure they are available and in use.
- Ensure patient comfort – consider appropriate pain management (use non-verbal aids if necessary).
- Maintain adequate hydration – particularly important in hypoactive delirium.
- Maintain a conducive environment -
    - Reduce noise to a minimum at night to promote sleep.
    - Avoid moving the patient within the unit, if at all possible.
    - Enhance orientation with e.g. lighting, signage, clocks etc
- Stimulate the patient during the day – organise interesting activities and involve family and friends.
- Avoid limiting the patient to bed -
    - Avoid unnecessary lines and catheters
    - Mobilise the patients early
- Avoid Benzodiazepines and ‘Z’ drugs.

Pharmacological Management of Delirium

There is no role for pharmacological treatment of hypoactive delirium, and treatment must focus on non-pharmacological treatment. However, if sleep is disrupted, melatonin can be used. Atypical anti-psychotics such as trazadone 50-100mg PO/NG can be used second line, but should be stopped when sleep pattern normalises and before ICU discharge.

In hyper-active delirium, there is no role for routine use of haloperidol. Typical (haloperidol) and atypical (such as trazadone & quetiapine) anti-psychotics should be reserved for when a patient has significant distress or is a risk to self or others. They do not reduce the duration of delirium or sequelae.

EMERGENCY TREATMENT	SUBACUTE TREATMENT	SPECIAL CIRCUMSTANCES
ATTEMPT TO DEESCALATE THE SITUATION: - Ensure patient safety and the safety of others - Get patients attention before speaking - Don't argue - Acknowledge their distress - Give rational explanations **PSYCHOTROPIC MEDICATION:** **HALOPERIDOL** - Check QTc <450 - 1-5mg IV Bolus - If no effect in 10 mins give same or increased dose (up to double previous dose), up to a max of 10mg per bolus. - Repeat as necessary at 10 min intervals using the 'double dose' regime until the patient is calm **or max 18mg/24hrs**[2] Or **CLONIDINE** - 50 – 150 micrograms IV or NG stat - Consider re-sedating and commencing a clonidine infusion 0-5mls/hr (750 micrograms in 50mls) before next sedation break. ††	**ONGOING CHALLENGING BEHAVIOUR:** Choose from the following options: **OLANZIPINE** - 5mg OD po **HALIPERIDOL** - 0.5 – 5mg OD po/ng **CLONIDINE** - 50-150mg IV or NG stat - Consider infusion 0-5mls/hr †† - Maximum dose 750 micrograms/24 hrs)	**NIGHT SEDATION** * caution with interactions if already taking pscychotropic medications* Consider: **MELATONIN** - 2mg od one hour prior to sleep time **With adjunct:** Option 1: **TRAZADONE** 50mg po nocte (7/7) Option 2: **AMYTRIPTILINE** 25mg po nocte Option 3: **HALOPERIDOL** 2-5mg IV nocte **WITHDRAWAL** **ALCOHOL –** follow local guidelines **OPIATE/ BENZODIAZEPINES –** Avoid abrupt withdrawal. Consult substance misuse team. If S&S of withdrawal restart and taper dose over longer. Consider long acting agents eg Methadone. Consider clonidine

Step 4: Immobility

Up to 50% of ICU survivors have ICU acquired muscle weakness and rehab is key to outcomes after a critical illness. Immobility worsens delirium (specifically restraint) and worsens muscle loss whilst delaying physical rehabilitation. The Medical Research Council *Sum Score* should be used to diagnose critical illness polymyoneuropathy, with a cut off of 48. With no effective treatments other than physiotherapy, prevention is key. Modifiable risk factors for medical staff to optimise are avoidance of use of neuromuscular blockers, hyper-osmolar states, hyperglycaemia and deranged calcium levels.[4]

Early mobilisation is the goal on our unit, and is demonstrably safe[1]. Passive physical rehab can begin on day one, with active mobilisation beginning as soon as respiratory and cardiovascular stability and neurological status allows. This is not precluded by the use of respiratory support or vasoactive infusions, so long as the patient is stable on these. If the patient develops new respiratory or cardiovascular instability, or change in neurological state, the rehab should stop and rehab plan be reviewed.

Step 5: Sleep

Pain, environmental stimuli, healthcare-related interruptions, psychologic factors, respiratory factors, and medications each affect sleep quality in the ICU.

Use of assist-control ventilation at night (vs pressure support ventilation) assists sleep. Noise and light reduction are important to promote sleep. Please cluster all interventions with nurses, to avoid repeated awakenings for patients.

To assist sleep, first offer ear plugs and an eyeshades. Consider melatonin if evidence of night time wakefulness and daytime somnolescence. Consider trazadone for night sedation, but avoid benzodiazepines and 'Z' drugs such as zopiclone. Propofol should not be used to encourage sleep.

References:

1. Clinical Practice Guidelines for the Prevention and Management : Critical Care Medicine (lww.com)
2. Oxford Handbook of Anaesthesia, Third edition. Allman K.G. & Wilson I.H. (2012) Oxford University Press, UK. P.288
3. Aubanel S. Bruiset F. (2020). Therapeutic options for agitation in the intensive care unit. Anaesthesia and Critical Care Pain Med. 39(2020)pp.639-646.
4. Critical illness polyneuropathy and myopathy: a systematic review - PMC (nih.gov)

Appendix – scoring systems:

Critical Care Pain Observation Tool

Indicator	Description	Score	
Facial expression	No muscular tension observed	Relaxed, neutral	0
	Presence of frowning, brow lowering, orbit tightening, and levator contraction	Tense	1
	All of the above facial movements plus eyelid tightly closed	Grimacing	2
Body movements	Does not move at all (does not necessarily mean absence of pain)	Absence of movements	0
	Slow, cautious movements, touching or rubbing the pain site, seeking attention through movements	Protection	1
	Pulling tube, attempting to sit up, moving limbs/ thrashing, not following commands, striking at staff, trying to climb out of bed	Restlessness	2
Muscle tension Evaluation by passive flexion and extension of upper extremities	No resistance to passive movements	Relaxed	0
	Resistance to passive movements	Tense, rigid	1
	Strong resistance to passive movements, inability to complete them	Very tense or rigid	2
Compliance with the ventilator (intubated patients)	Alarms not activated, easy ventilation	Tolerating ventilator or movement	0
	Alarms stop spontaneously	Coughing but tolerating	1
	Asynchrony: blocking ventilation, alarms frequently activated	Fighting ventilator	2
OR			
Vocalization (extubated patients)	Talking in normal tone or no sound	Talking in normal tone or no sound	0
	Sighing, moaning	Sighing, moaning	1
	Crying out, sobbing	Crying out, sobbing	2

Richmond Agitation and Sedation Scale (RASS)		
+4	Combative	violent, immediate danger to staff
+3	Very Agitated	Pulls or removes tube(s) or catheter(s); aggressive
+2	Agitated	Frequent non-purposeful movement, fights ventilator
+1	Restless	Anxious, apprehensive but movements not aggressive or vigorous
0	Alert & calm	
-1	Drowsy	Not fully alert, but has sustained awakening to *voice* (eye opening & contact ≥ 10 sec)
-2	Light sedation	Briefly awakens to *voice* (eye opening & contact < 10 sec)
-3	Moderate sedation	Movement or eye-opening to *voice* (but no eye contact)
-4	Deep sedation	No response to voice, but movement or eye opening to *physical* stimulation
-5	Unarousable	No response to *voice or physical* stimulation

MRC Sum Score:

Movement tested on each side	Score for each movement
Arm abduction	0=no movement
Flexion at the elbow	1=flicker of movement
Wrist extension	2=movement with gravity eliminated
Hip flexion	3=movement against gravity
Extension at the knee	4=movement against resistance
Ankle dorsiflexion	5=normal power